

# Discussions Feature

▼ Relevant source files

Frontend/STUDER/src/app/features/users/user.model.ts

Frontend/STUDER/src/app/features/users/user.service.ts

Frontend/STUDER/src/assets/i18n/en.json

Frontend/STUDER/src/assets/i18n/es.json

The **Discussions Feature** provides a collaborative space for users to engage in public or private topical threads. It supports threaded messaging, tag-based filtering, favoriting, and activity-based ordering. The feature is implemented with a strict distinction between open and closed states, where closed discussions transition to a read-only mode.

## Data Flow and Architecture

The feature follows the standard STUDER frontend pattern: a centralized service (`DiscussionService`) manages API interactions, while components handle the UI logic for listing and detailed viewing.

### Discussion Entity Space

The system uses several DTOs to represent the state of discussions and messages:

| Entity / DTO                              | Role                                                                                         | Source                                                                 |
|-------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <code>DiscussionResponseDTO</code>        | Represents the metadata of a discussion thread (title, tags, creator, state).                | Frontend/STUDER/src/app/features/discussions/discussion.model.ts 18-31 |
| <code>DiscussionMessageResponseDTO</code> | Represents a single message within a thread, including likes and parent/child relationships. | Frontend/STUDER/src/app/features/discussions/discussion.model.ts 42-53 |
| <code>DiscussionCreateRequestDTO</code>   | Payload for creating a new thread.                                                           | Frontend/STUDER/src/app/features/discussions/discussion.model.ts 12-16 |

| Entity / DTO                   | Role                                                      | Source                                                                 |
|--------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|
| <b>MessageCreateRequestDTO</b> | Payload for posting a reply or a new message in a thread. | Frontend/STUDER/src/app/features/discussions/discussion.model.ts 37-40 |

**Sources:** Frontend/STUDER/src/app/features/discussions/discussion.model.ts 1-60

## DiscussionService Implementation

The **DiscussionService** acts as the primary interface for the backend REST API. It handles pagination, filtering, and state updates (like favoriting).

### Key Functions

- getPublicDiscussions(page, size, tags)**: Fetches discussions visible to all users. Supports filtering by an array of tags
  
Frontend/STUDER/src/app/features/discussions/discussion.service.ts 21-25
- getMyDiscussions(page, size)**: Retrieves threads created by the authenticated user
  
Frontend/STUDER/src/app/features/discussions/discussion.service.ts 27-29
- create(request)**: POSTs a new discussion to the **/api/v1/discussions** endpoint
  
Frontend/STUDER/src/app/features/discussions/discussion.service.ts 35-37
- getMessages(discussionId, page, size)**: Retrieves the threaded messages for a specific discussion
  
Frontend/STUDER/src/app/features/discussions/discussion.service.ts 43-45
- close(id)**: Transitions a discussion to the closed state, preventing further messages
  
Frontend/STUDER/src/app/features/discussions/discussion.service.ts 63-65

**Sources:** Frontend/STUDER/src/app/features/discussions/discussion.service.ts 1-66

## UI Components

The feature is divided into two main views: the list view and the detailed thread view.

### DiscussionListComponent

This component displays a paginated list of threads.

- Filtering**: Users can toggle between "Public" and "My Discussions"
  
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- **Tag Filtering:** Discussions can be filtered by specific tags, which are passed as query parameters to the `DiscussionService`

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- **Ordering:** Threads are ordered by "Activity Hour," ensuring that discussions with recent replies appear at the top of the list.

## DiscussionDetailComponent

This component manages the interaction within a specific thread.

- **Threaded Messaging:** Displays messages and allows for replies. It uses the `RichTextComponent` to render mentions and tags within messages.
- **Closed State:** If a discussion is marked as closed, the UI disables the message input and displays a "Closed" badge

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- **Favourites:** Users can add threads to their favorites via `addFavourite(id)`

Frontend/STUDER/src/app/features/discussions/discussion.service.ts 55-57

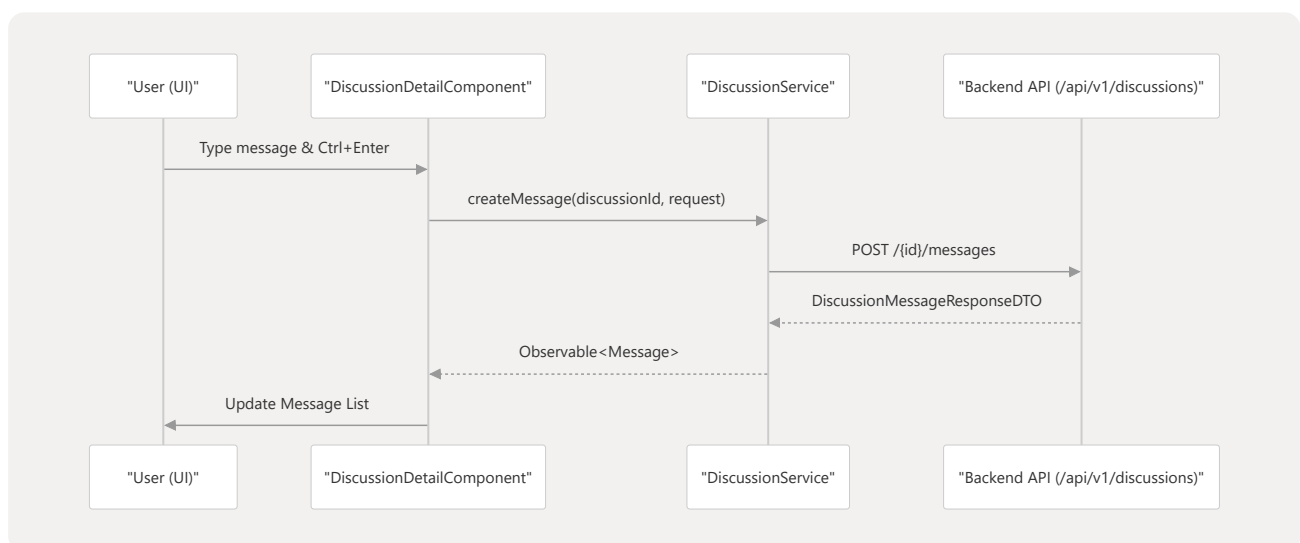
**Sources:** Frontend/STUDER/src/app/features/discussions/discussion.service.ts 1-66

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## Technical Diagrams

### Discussion Creation and Message Flow

The following diagram illustrates the interaction between the UI components and the `DiscussionService`.

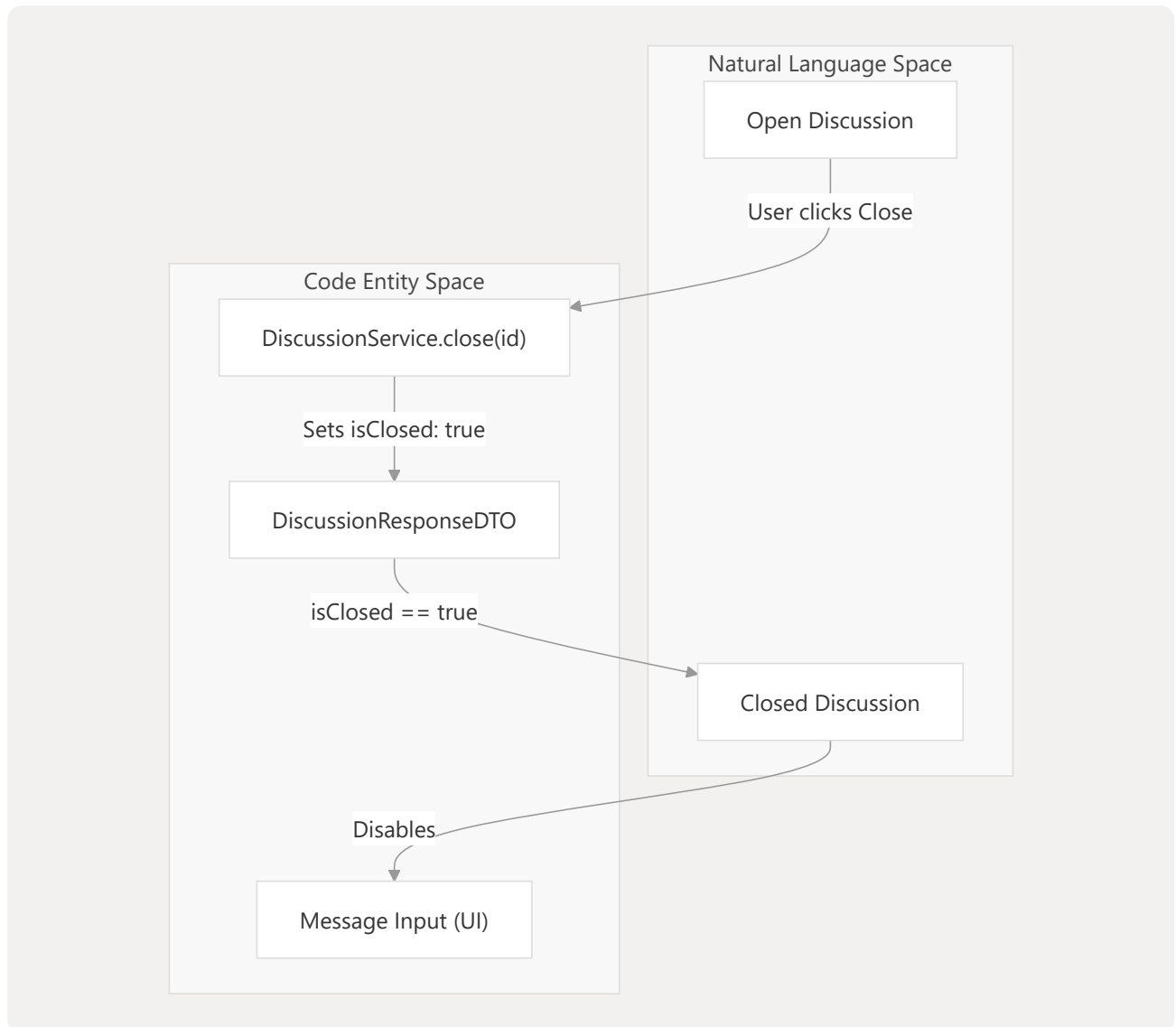


**Sources:** Frontend/STUDER/src/app/features/discussions/discussion.service.ts 47-49

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## State Management: Open vs Closed Discussions

This diagram maps the natural language "Closed State" to the code entities that enforce it.



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## Key Localization Keys

The feature relies on the following i18n keys for its interface:

- `discussions.create` : "New discussion" Frontend/STUDER/src/assets/i18n/en.json 150
- `discussions.replying_to` : "Replying to" Frontend/STUDER/src/assets/i18n/en.json 161
- `discussions.form_tags_placeholder` : "Type and press Enter..." Frontend/STUDER/src/assets/i18n/en.json 168

**Sources:** Frontend/STUDER/src/assets/i18n/en.json 148-173